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MA3521 — Variation of Parameters

Name:

Solve $\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t)$ $\quad A = \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix}$ $\quad \mathbf{f}(t) = \begin{pmatrix} e^t \sec(3t) \\ 0 \end{pmatrix}$

$\lambda = \alpha + \beta i = 1 + 3i \quad \Longrightarrow \quad \alpha = 1,\ \beta = 3 \qquad \mathbf{V} = \mathbf{A} + i\mathbf{B} = \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

$\mathbf{Y}_1 = e^{\alpha t}\big(\mathbf{A}\cos(\beta t) - \mathbf{B}\sin(\beta t)\big) = e^t \begin{pmatrix} \cos(3t) \\ \sin(3t) \end{pmatrix}$

$\mathbf{Y}_2 = e^{\alpha t}\big(\mathbf{A}\sin(\beta t) + \mathbf{B}\cos(\beta t)\big) = e^t \begin{pmatrix} \sin(3t) \\ -\cos(3t) \end{pmatrix} \qquad \Phi = [\,\mathbf{Y}_1 \quad -\mathbf{Y}_2\,]$

1. $\mathbf{y}_c =$ $e^t\left[c_1 \begin{pmatrix} \cos(3t) \\ \sin(3t) \end{pmatrix} + c_2 \begin{pmatrix} -\sin(3t) \\ \cos(3t) \end{pmatrix}\right]$ $\qquad \Phi(t) =$ $e^t \begin{pmatrix} \cos(3t) & -\sin(3t) \\ \sin(3t) & \cos(3t) \end{pmatrix}$

2. $\det \Phi(t) =$ e^{2t} $\qquad \Phi(t)^{-1} =$ $e^{-t} \begin{pmatrix} \cos(3t) & \sin(3t) \\ -\sin(3t) & \cos(3t) \end{pmatrix}$

$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad \det M = AD - BC$

$M^{-1} = \dfrac{1}{AD - BC} \begin{pmatrix} D & -B \\ -C & A \end{pmatrix}$

$AD - BC \neq 0$

3. $\mathbf{u}'(t) =$ $\Phi(t)^{-1}\mathbf{f}(t)$

$\Phi^{-1}\mathbf{f} = e^{-t} \begin{pmatrix} \cos(3t) & \sin(3t) \\ -\sin(3t) & \cos(3t) \end{pmatrix} \begin{pmatrix} e^t \sec(3t) \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -\tan(3t) \end{pmatrix} \qquad \mathbf{u} = \begin{pmatrix} t \\ \frac{1}{3}\ln|\cos(3t)| \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$\boxed{\mathbf{y} = \mathbf{y}_c + e^t \begin{pmatrix} t\cos(3t) - \frac{1}{3}\sin(3t)\ln|\cos(3t)| \\ t\sin(3t) + \frac{1}{3}\cos(3t)\ln|\cos(3t)| \end{pmatrix}} \quad \cos(3t) \neq 0$

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Solve $\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t) \quad A = \begin{pmatrix} -5 & 3 \\ -8 & 5 \end{pmatrix} \quad \mathbf{f}(t) = \dfrac{e^t}{1+t^2}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$\lambda_1 = 1, \quad \mathbf{V}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \qquad \lambda_2 = -1, \quad \mathbf{V}_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$\mathbf{Y}_1 = e^t\mathbf{V}_1, \quad \mathbf{Y}_2 = e^{-t}\mathbf{V}_2, \quad \Phi = [\,\mathbf{Y}_1 \;\; \mathbf{Y}_2\,], \quad \mathbf{y}_c = c_1\mathbf{Y}_1 + c_2\mathbf{Y}_2$

1. $\mathbf{y}_c =$ $c_1 e^t \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ $\qquad \Phi(t) =$ $\begin{pmatrix} e^t & 3e^{-t} \\ 2e^t & 4e^{-t} \end{pmatrix}$

2. $\det \Phi(t) =$ -2 $\qquad \Phi(t)^{-1} =$ $-\frac{1}{2}\begin{pmatrix} 4e^{-t} & -3e^{-t} \\ -2e^t & e^t \end{pmatrix}$

3. $\mathbf{u}'(t) =$ $\Phi(t)^{-1}\mathbf{f}(t)$

$\Phi^{-1}\mathbf{f} = -\dfrac{1}{2}\begin{pmatrix} 4e^{-t} & -3e^{-t} \\ -2e^t & e^t \end{pmatrix} \dfrac{e^t}{1+t^2}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = -\dfrac{1}{2(1+t^2)}\begin{pmatrix} 4-6 \\ -2e^{2t}+2e^{2t} \end{pmatrix} = \begin{pmatrix} 1/(1+t^2) \\ 0 \end{pmatrix}$

$\mathbf{u} = \begin{pmatrix} \arctan(t) \\ 0 \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$$\boxed{\mathbf{y} = e^t(c_1 + \arctan(t))\begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} 3 \\ 4 \end{pmatrix}}$$